
Corrigé — Exercice 12 — Bac principal 2024 (partie suite)

1) $u_1 = 4 \cdot 1 - 1 = 3$, $u_2 = 4 \cdot 3 + 12 - 1 = 23$, $u_3 = 4 \cdot 23 + 24 - 1 = 115$.

2a) $v_{n+1} = u_{n+1} + 4(n+1) + 1 = (4u_n + 12n - 1) + 4n + 5 = 4u_n + 16n + 4 = 4(u_n + 4n + 1) = 4v_n$:
géométrique de raison 4.

2b) $v_0 = u_0 + 1 = 2$, donc $v_n = 2 \cdot 4^n$.

2c) $u_n = v_n - 4n - 1 = 2 \cdot 4^n - 4n - 1$.

3) Le terme $2 \cdot 4^n$ l'emporte, donc $\lim_{n \rightarrow +\infty} u_n = +\infty$.